HUI Tools

User Manual

HUI-Display • HUI-Test

A software toolkit for the Mackie HUI controller

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Important Notice

⚠ **WARNING:** HUI-Display and HUI-Test are independent programs. Running both simultaneously while Pro Tools is connected can cause erratic fader movement, display corruption, and loss of the Pro Tools–HUI connection. See Chapter 4 for details.

⚠ **WARNING:** HUI is a product of Mackie Designs. This software is an unofficial third-party utility and is not affiliated with, endorsed by, or supported by Mackie Designs or Avid Technology. Use at your own risk.

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Contents

1 Introduction

1.1 What is HUI Tools?
1.2 The Mackie HUI
1.3 Package Contents

2 Installation

2.1 Step 1 — Install Python
2.2 Step 2 — Install loopMIDI
2.3 Step 3 — Install Python Libraries
2.4 Finding Your HUI Port Names

3 HUI-Display

3.1 Overview
3.2 First Launch and Configuration
3.3 Configuring Pro Tools
3.4 Daily Use
3.5 The About Dialog

4 HUI-Test

4.1 Critical Warning
4.2 Overview
4.3 Configuration
4.4 Connection Tab
4.5 Live View Tab
4.6 Display Tab
4.7 LEDs Tab
4.8 Meters and V-Pots Tab
4.9 Faders Tab
4.10 Audio and Relays Tab
4.11 The VFD Display Window

5 The config.hui File

6 Troubleshooting

7 Limitations

8 Licensing

Chapter 1

Introduction

1.1 What is HUI Tools?

HUI Tools is a small suite of software utilities for the Mackie HUI (Human User Interface) studio controller. It was created to solve a specific problem: the vacuum-fluorescent display (VFD) on a HUI that has failed, leaving the user unable to see plug-in parameters, dialog boxes, or other text that Pro Tools sends to that screen. HUI Tools intercepts and re-displays that information on a computer monitor, and also provides a comprehensive hardware test tool for diagnosing HUI problems.

The suite consists of two programs:

- **HUI-Display** — Intercepts the MIDI data that Pro Tools sends to the HUI's main display, shows it in a floating on-screen window, and passes all traffic through to the hardware unchanged so the rest of the unit continues to work normally.
- **HUI-Test** — Connects directly to the HUI hardware and lets you individually test every feature of the unit: LEDs, faders, V-pot rings, VU meters, the VFD display, relays, and the click and beep circuits.

1.2 The Mackie HUI

The Mackie HUI is a MIDI-controlled studio console surface manufactured by Mackie Designs in 1997. It communicates with Pro Tools (and compatible DAWs) via a MIDI connection, carrying fader positions, button presses, LED states, and display data back and forth. The main display is a 2-row, 40-character vacuum-fluorescent display (VFD) located in the meter bridge.

HUI Tools communicates with the HUI using the same MIDI protocol that Pro Tools uses, documented in the HUI MIDI Protocol specification by theageman (2010).

1.3 Package Contents

The HUI Tools package contains the following files. All files should be kept in the same folder.

File	Description
hui_display.py	The HUI-Display program. Run this during Pro Tools sessions.
hui_test.py	The HUI-Test program. Run this for hardware diagnostics.
list_midi_ports.py	A small helper utility that lists all MIDI devices on your system.
config.hui	Shared settings file, created automatically on first launch.

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SETUP_INSTRUCTIONS.txt	Quick-reference setup guide.

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Chapter 2

Installation

You only need to complete this installation once. After that, running either program is just a double-click.

> **NOTE:** HUI-Test does not require loopMIDI. If you only intend to use HUI-Test, you can skip Step 2.

2.1 Step 1 — Install Python

Both programs are written in Python and require it to be installed on your computer.

1. Go to https://www.python.org/downloads/ and click the large Download Python button.
2. Run the installer.
3. **CRITICAL:** On the first screen of the installer, tick the checkbox labelled **"Add Python to PATH"** before clicking Install Now. If you skip this, the programs will not run.
4. Click Install Now and wait for it to complete.

2.2 Step 2 — Install loopMIDI (HUI-Display only)

HUI-Display needs virtual MIDI ports to sit between Pro Tools and the HUI hardware. loopMIDI is a free Windows utility that creates these ports.

5. Go to https://www.tobias-erichsen.de/software/loopmidi.html and download loopMIDI.
6. Run the installer.
7. Launch loopMIDI. It will appear as a small icon in the Windows system tray (bottom-right corner of the screen).
8. In the text box at the bottom of the loopMIDI window, type `HUI In` and click the + button. A new port will appear in the list.
9. Repeat: type `HUI Out` and click + again. You should now see both ports in the list.

> ✓ **TIP:** In loopMIDI's Options menu, enable "Autostart with Windows" so the virtual ports are available every time you start your computer without needing to open loopMIDI manually.

2.3 Step 3 — Install Python Libraries

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Both programs rely on two Python libraries: mido (MIDI handling) and python-rtmidi (communication with your MIDI hardware). Install them using the Command Prompt.

Quick method

10. Open Command Prompt. Press the Windows key, type `cmd`, and press Enter.
11. Run the following command to update the installer:

```
python -m pip install --upgrade pip
```

12. Then run:

```
pip install mido python-rtmidi
```

13. Wait for the installation to complete. You may see a message about "scripts not on PATH" — this is harmless and can be ignored.

If the installation fails

If you see an error message containing the words "Unknown compiler" or "Visual Studio", python-rtmidi needs a C++ compiler that is not installed on your system. Follow these steps:

14. Go to https://visualstudio.microsoft.com/visual-cpp-build-tools/ and click Download Build Tools.
15. Run the installer. When it asks what to install, tick **"Desktop development with C++"** and click Install. The download is approximately 1–2 GB.
16. Restart your computer.
17. Open Command Prompt again and re-run:

```
pip install mido python-rtmidi
```

2.4 Finding Your HUI Port Names

Windows appends a number to every MIDI port name (for example, your Roland UM-ONE interface might appear as "UM-ONE 2" in the input list and "UM-ONE 3" in the output list). You need these exact names to configure the programs.

To see all available MIDI ports on your system, run the helper script:

```
python list_midi_ports.py
```

This will display two lists — input ports and output ports — with the full name of each device including the trailing number. Write down the names of your HUI hardware's input and output ports.

> **NOTE:** Port numbers can change if you plug or unplug USB devices, or change the order in which they are powered on. If a program stops connecting after you have changed your USB setup, run list_midi_ports.py again to find the new numbers.

Chapter 3

HUI-Display

3.1 Overview

HUI-Display sits invisibly between Pro Tools and your HUI hardware, forwarding all MIDI traffic in both directions while watching for the specific SysEx messages that update the main 2×40 character VFD display. When those messages arrive, it decodes the text and shows it in a floating on-screen window styled to resemble the original display.

The signal flow looks like this:

```
Pro Tools  →  [HUI In  (loopMIDI)]  →  HUI-Display  →  HUI hardware
HUI hardware  →  HUI-Display  →  [HUI Out (loopMIDI)]  →  Pro Tools
```

Every MIDI message passes through unchanged, so your faders, buttons, and all other HUI functions continue to work exactly as before. The only difference is that the on-screen window now shows what the broken VFD would have shown.

3.2 First Launch and Configuration

On first launch, HUI-Display creates a config.hui settings file in the same folder as the program. You then configure your MIDI ports from within the program itself.

18. Make sure your HUI hardware is turned on and connected.
19. Make sure loopMIDI is running with the HUI In and HUI Out ports created (see Chapter 2).
20. Double-click `hui_display.py` to launch the program. The display window will appear.
21. Click Menu → Configure in the menu bar.

MIDI Ports tab

This tab has four drop-down menus. Set them as follows:

Port	What to select
Pro Tools → Script (input)	Select HUI In (the loopMIDI virtual port, with its number suffix)
Script → HUI hardware (output)	Select your physical HUI MIDI interface (e.g. "UM-ONE 3")
HUI hardware → Script (input)	Select your physical HUI MIDI interface (e.g. "UM-ONE 2")
Script → Pro Tools (output)	Select HUI Out (the loopMIDI virtual port, with its number suffix)

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> **NOTE:** If your device does not appear in a drop-down, click the Refresh port list button to re-scan.

> **NOTE:** On Windows, the same physical device often appears with different numbers in the input and output lists. Make sure you select the correct list entry for each field — the "HUI hardware input" field shows input ports, and the "HUI hardware output" field shows output ports.

Appearance tab

This tab lets you change the display text colour, background colour, and font size to suit your preferences. Click any colour swatch or the Choose button to open the Windows colour picker. The Reset visual settings to defaults button restores the original teal-on-black VFD aesthetic.

Click Apply & Reconnect when you are done. Your settings are saved to config.hui automatically.

3.3 Configuring Pro Tools

You need to tell Pro Tools to route its HUI traffic through the loopMIDI virtual ports instead of directly to the hardware.

22. In Pro Tools, go to Setup → Peripherals → MIDI Controllers tab.
23. Find your HUI entry (Type = HUI).
24. Change the two port settings as follows:

> **WARNING:** The terms "Send To" and "Receive From" are from Pro Tools' perspective — the opposite of what you might expect. **"Send To" is where Pro Tools sends data outward (to the HUI). "Receive From" is where Pro Tools listens for data coming back.**

Pro Tools setting	What to enter
Send To	HUI In ← Pro Tools outputs here; the script reads it
Receive From	HUI Out ← The script writes here; Pro Tools reads it

25. Click OK. The HUI should come online within a few seconds.

3.4 Daily Use

Each time you want to use HUI-Display:

26. Start loopMIDI (or let it start automatically with Windows).
27. Turn on your HUI hardware.
28. Double-click `hui_display.py`. The display window appears.
29. Open Pro Tools as normal. As soon as a session loads, the on-screen display will populate with channel names, parameter values, and plug-in dialogs just as the original VFD would have done.

The window can be dragged anywhere on your screen by clicking and dragging. It always stays on top of other windows. The status bar at the bottom shows how many messages have been received, including a count of display SysEx messages — if this number stays at zero while Pro Tools is running, check that loopMIDI is running and that the Pro Tools configuration in step 3.3 is correct.

3.5 The About Dialog

Click About in the menu bar to view the program version, authorship, and licensing information, including a clickable link to the full licence text.

Chapter 4

HUI-Test

4.1 Critical Warning

⚠ **WARNING: DO NOT run HUI-Test at the same time as HUI-Display with Pro Tools connected.** HUI-Test connects directly to your HUI hardware over MIDI. If HUI-Display is simultaneously routing Pro Tools traffic through the same ports, both programs will compete for access. This can cause erratic and uncontrolled fader movement, corrupted display output, Pro Tools losing its HUI connection, and loopMIDI auto-muting the virtual ports. Before running HUI-Test, either close HUI-Display, or disconnect the HUI in Pro Tools (Setup → Peripherals → MIDI Controllers).

A warning dialog reminding you of this is shown each time the program launches. A "Do not show this warning again" checkbox is available once you are familiar with the requirement.

4.2 Overview

HUI-Test is a diagnostic tool that communicates directly with the HUI hardware. It lets you independently test every aspect of the unit without needing Pro Tools to be connected. This is useful for verifying hardware operation, testing repairs, or learning how the HUI protocol works.

Unlike HUI-Display, HUI-Test does NOT require loopMIDI. It connects directly to your HUI hardware using the same MIDI port names you use in HUI-Display.

4.3 Configuration

30. Double-click `hui_test.py`. The startup warning dialog will appear. Dismiss it.
31. Click Menu → Configure.
32. Set the two port fields to your physical HUI hardware ports (the same port names you used in HUI-Display's "Script → HUI hardware" and "HUI hardware → Script" fields).
33. Click Apply.
34. Click the Connect button. The button will turn red and show "Disconnect" when connected. A welcome message will appear briefly on the HUI VFD.

4.4 Connection Tab

The Connection tab is the control centre for HUI-Test.

Ping

The HUI protocol uses an active-sensing "ping" to confirm that MIDI communication is active. Click Send Ping to send a ping message (note on, channel 0, note 0, velocity 0). If the HUI is properly connected, it will respond with a reply (note on, velocity 127) within a few milliseconds, which the program will display next to the Last reply indicator.

Auto-ping

The Auto-ping checkbox (enabled by default) causes the program to send a ping automatically every second. This keeps the HUI in online mode — without regular pings, the HUI will go offline after approximately two seconds, which stops the motorised faders from responding. Keep auto-ping enabled during testing.

MIDI Receive Log

Every MIDI message received from the HUI hardware appears in this log in real time. This includes button presses, fader touch events, fader position changes, V-pot turns, and jog wheel movement. This is useful for verifying that every control on the surface is transmitting correctly.

4.5 Live View Tab

The Live View tab provides a real-time graphical display of everything the HUI hardware is sending back to the computer. It is the primary test tool for verifying that the physical controls on the surface are working correctly — press a button and watch it light up on screen; move a fader and see the bar rise; turn a V-pot and see the indicator dot move. Any control that fails to produce a visible response in the Live View is not transmitting MIDI data.

> **NOTE:** The Live View shows data travelling from the HUI hardware to the computer (hardware → host direction). It does not show the LED, meter, or display states that HUI-Test is sending in the other direction. Those are controlled in the LEDs, Meters & V-Pots, and Display tabs.

Graphical Canvas

The canvas fills the window and scales automatically when you resize it. The layout is divided into four sections from left to right, mirroring the physical panel arrangement of the HUI:

Section	Contents
Left control strip	Buttons for the SELECT-ASSIGN section (SEND A–E, PAN, MUTE, SHIFT), assignment controls (SUSPEND, DEFAULT, ASSIGN, OUTPUT, INPUT), and channel/bank scroll (←CH, ←BK, CH→, BK→)

8 channel strips	One column per channel. Each shows a fader touch indicator (red dot when the fader cap is being touched), a V-pot ring (11 dots in a 270° arc with a single bright dot tracking the accumulated encoder position), LED buttons for REC/RDY, INSERT, V-SEL, AUTO, SOLO, MUTE, and SELECT, and a fader bar showing the current fader position as a percentage.
Transport section	Rows of transport buttons (RWND, FFWD, STOP, PLAY, REC; RTZ, END, ONLINE, LOOP, QUICK PUNCH; AUDITION, PRE, IN, OUT, POST; TALKBACK, MODE, SCRUB, SHUTTLE), a rotating-needle jog wheel indicator, and a ping flash indicator.
Right panel	All remaining zones shown as compact labelled groups: KB Shortcuts, Window, DSP Param, Auto Enable, Auto Mode, Status/Group, Edit, Function Keys, Monitor In, Monitor Out, and Timecode LEDs.

When a button on the HUI is physically pressed, the corresponding element in the canvas lights up in bright teal with inverted text. It returns to dim when the button is released. Fader bars fill upward in real time as faders are moved. The V-pot ring indicator dot accumulates clockwise and counter-clockwise turns from a neutral centre position; it resets to centre on connect.

Decoded Log

A scrollable log below the canvas shows every incoming MIDI event in plain English. Examples: "Ch 3 Fader → 73%", "Auto Enable · fader → ▼ pressed", "Jog Wheel → CW ×2 (total: 47)". The log appends new entries incrementally and never blocks the graphical display.

Two checkboxes control the log:

- **Decoded log** — Shows or hides the log panel entirely.
- **Show raw MIDI** — When enabled, adds an indented second line beneath each decoded entry showing the raw mido MIDI message string. Useful when comparing against the HUI protocol specification.

The Clear button resets the displayed log. The raw MIDI toggle rebuilds the log display from the current history when switched.

4.6 Display Tab

The Display tab lets you send text to all of the HUI's display elements.

Main VFD (2×40 display)

The two text boxes correspond to the two rows of the main 40-character display. Each box accepts a maximum of 40 characters. Type your text and click Send to VFD. Clear VFD clears both rows. Test pattern fills the display with a repeating 1–0 / A–J sequence to check that all character positions are working.

The VFD uses a specific character set (HUI-2) that is slightly different from standard ASCII. Most printable characters are the same, but a few special symbols (arrows, block graphics, note symbols, degree signs) are available in the lower range. The program handles this character set automatically.

Channel Scribble Strips

The HUI has nine 4-character LED dot-matrix displays above the fader section — one for each of the eight channel strips, plus the SELECT-ASSIGN display. Each text box accepts a maximum of 4 characters. Click Send All to update all nine displays simultaneously.

Timecode Display

The timecode display is a row of eight 7-segment LED digits showing the current position in hours, minutes, seconds, and frames (HH:MM:SS:FF). Enter your timecode value in the box and click Send. The separating dots between fields are automatically encoded by the program — you do not need to add them manually.

Timecode Counter

Below the send controls is an animated counter. Enter a starting timecode value in the HH:MM:SS:FF field above, set the desired frame rate in the FPS box (any integer from 1 to 240; the HUI does not enforce a specific rate), then click Start Counter. The counter increments the timecode by one frame on each tick, sends the corresponding SysEx command automatically, and handles all field carry-overs (frames → seconds → minutes → hours). Click Stop Counter to halt it. The timecode display on the hardware will continue to show the last value sent.

4.7 LEDs Tab

The HUI's LED system is organised into 29 zones, each containing up to 8 ports. Every LED has a zone number and a port number within that zone, and the same address is used for the corresponding button.

Zone list

The list on the left shows all zones that contain LEDs. Zones that contain only buttons with no LED (the numeric keypad) and the relay/click/beep zone (tested in the Audio tab) are not shown here.

Port buttons

When you select a zone from the list, the right side shows the ports in that zone as toggleable buttons. Each button displays the function name of the corresponding LED (e.g. "mute", "solo",

"record"). Click a button to toggle the LED on or off. An illuminated button indicates the LED is currently on. Ports that exist as buttons only (no LED) are not shown.

> **NOTE: Zone 0x0D (Cursor / Mode / Scrub):** Only the Mode, Scrub, and Shuttle buttons have LEDs. The Up, Down, Left, and Right arrow buttons are plain switches with no LED and do not appear in this panel.

All-On and All-Off buttons

"Zone: All Ports ON/OFF" switches all LED ports in the currently selected zone at once. "All LEDs ON/OFF (all zones)" does the same for every zone in the list simultaneously, which is useful for a quick visual check of whether all LEDs on the surface are functioning.

4.8 Meters and V-Pots Tab

VU Meters

The HUI has two LED ladder meters (left and right) above each of the eight channel strips, with twelve segments each. The sliders in this tab set the level for each meter. The scale runs from 0 (all LEDs off) to 12 (all LEDs lit, including the red clip LED).

The Sweep toggle button runs a continuous animation across all meters simultaneously. While active, each meter follows a sine wave that rises from 0 to 12 and back down. The Speed slider (0.2× to 5×) controls how fast the wave cycles — at 1×, a full cycle takes about six seconds. Click the button again to stop; all meters will silence when the sweep ends.

V-Pot Rings

Each V-pot (the rotary encoders) is surrounded by a ring of 11 LEDs. The ring can display values in four different modes:

Mode	Description
Dot	A single LED moves across the ring. Used for absolute position display.
Pan	LEDs fill from the centre outward, left and right. Used for pan/balance.
Level	LEDs fill from the left edge. Used for send levels and similar parameters.
Spread	LEDs expand symmetrically from the centre. Used for width or spread controls.

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The position slider (0–11) controls where in the ring the value appears. The Centre LED checkbox independently controls a small indicator LED mounted directly beneath the encoder knob.

The V-Pot Sweep toggle button animates all 13 V-pot rings simultaneously in Dot mode. Each pot is given a slightly different phase offset so they form a rolling wave across the surface rather than moving in lockstep. The Speed slider controls the animation rate, identical to the VU meter sweep. Click the button again to stop; all rings will be silenced.

4.9 Faders Tab

Manual control

The eight vertical sliders drive the motorised faders directly. Drag a slider to move the corresponding fader. The percentage value is shown above each slider. The All to Top, All to Centre, and All to Bottom buttons move all faders to those positions simultaneously.

> ⚠ **WARNING:** Moving faders remotely via software will cause them to move physically. Ensure that nothing is resting on or caught under any fader knob before sending fader commands.

Automated Motion Demo

The demo section animates all eight faders automatically in a repeating pattern. Click Start Demo to begin; click Stop Demo to end.

Five wave shapes are available:

Shape	Description
Sine	Smooth, continuous oscillation. The most natural-looking motion.
Square	Rapid jumps between high and low. Tests motor speed and accuracy.
Triangle	Linear ramp up, then linear ramp down. Consistent speed throughout.
Sawtooth	Ramps steadily up, then instantly resets to the bottom.
Bounce	Simulates a ball bouncing — fast through the middle, decelerating at the top.

Three sliders control the motion:

- **Speed** — How many cycles per second. At 1.0×, one complete wave takes approximately two seconds. Range: 0.2× (very slow) to 4.0× (rapid).
- **Amplitude** — How far the faders travel from the centre position. At 100%, the faders sweep the full range. At 20%, they make small movements around the centre.

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- **Centre** — The midpoint of the motion. At 50%, faders oscillate around the centre. At 80%, they oscillate in the upper region of travel.

The Phase spread checkbox offsets each successive fader by 45° so that instead of all moving together, they form a rolling wave pattern across the surface. When disabled, all faders move in perfect synchronisation.

4.10 Audio and Relays Tab

This tab controls the non-LED outputs in Zone 0x1D of the HUI.

Relays (Zone 1D, Ports 0 and 1)

The HUI has two output relays that can be used to trigger external circuits — for example, an illuminated "Recording in Progress" sign or an "On Air" light. Each relay is a simple normally-open contact that closes when activated. Click the toggle button to switch each relay on or off.

Click (Zone 1D, Port 2)

The HUI contains a small speaker that can produce a brief click sound, typically used to confirm button presses. Click the Single Click button to trigger it once. Use the Repeat field and Click N times button to trigger it multiple times in sequence. Unlike the beeper (below), the click fires once and stops automatically — there is no "off" state.

Beeper (Zone 1D, Port 3)

The HUI also contains a continuous beeper for indicating error conditions. Unlike the click, the beeper continues sounding until you explicitly turn it off. The toggle button shows its current state (ON or OFF). Always turn the beeper off when you are done testing it.

4.11 The VFD Display Window

Menu → View VFD Display opens a floating window that shows the current content of the main 2×40 VFD display, styled identically to the HUI-Display window. This reflects whatever text HUI-Test has most recently sent to the display — the welcome message on connect, any text you have typed in the Display tab, or test patterns.

> **NOTE:** This window shows what HUI-Test has sent, not what is currently being displayed on the physical VFD hardware. If you send text and then disconnect, the window will continue to show the last text that was sent.

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Chapter 5

The config.hui File

config.hui is a shared settings file used by both HUI-Display and HUI-Test. It is created automatically the first time either program is launched, and is stored in the same folder as the program files.

The file is in JSON format (plain text) and can be opened and read in Notepad if needed, though it is much easier to use the in-program Configure dialog. A typical config.hui file looks like this:

```
{
  "pt_sends_to":         "HUI In 3",
  "hui_out_port":        "UM-ONE 3",
  "hui_in_port":         "UM-ONE 2",
  "pt_recv_from":        "HUI Out 5",
  "fg_color":            "#00eed5",
  "bg_color":            "#06100e",
  "dim_color":           "#005549",
  "font_size":           15,
  "skip_test_warning":   false
}
```

Key	Meaning
pt_sends_to	loopMIDI port that Pro Tools sends to (HUI-Display only)
hui_out_port	Physical HUI hardware MIDI output port
hui_in_port	Physical HUI hardware MIDI input port
pt_recv_from	loopMIDI port that Pro Tools receives from (HUI-Display only)
fg_color	Display text colour (hex code)
bg_color	Window background colour (hex code)
dim_color	Label and border colour (hex code)
font_size	Display font size in points
skip_test_warning	If true, suppresses the HUI-Test startup warning

NOTE: If config.hui becomes corrupted or you want to start fresh, simply delete it. Both programs will create a new one with default settings on next launch.

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Chapter 6

Troubleshooting

"MIDI ERROR" appears in the status bar

- Open Menu → Configure → MIDI Ports. Click the Refresh port list button and re-select your ports from the updated list.
- Check that loopMIDI is running (HUI-Display only). Look for its icon in the Windows system tray.
- Check that no other program has exclusive access to the same MIDI ports.
- Run list_midi_ports.py to see the current names and numbers of all ports.

The display window shows blank text

- Make sure Pro Tools is running and a session is open. The VFD only receives data when Pro Tools is actively communicating with the HUI.
- Check the status bar — if the "display SysEx received" counter is at zero, no display data is reaching HUI-Display. Verify your Pro Tools peripheral settings (Chapter 3.3) and make sure you have Send To and Receive From set correctly.
- Try disconnecting and reconnecting the HUI in Pro Tools (Setup → Peripherals) to trigger a fresh display update.

loopMIDI has muted one or both virtual ports

loopMIDI includes automatic feedback detection. When it sees MIDI data arriving back on a port it just sent from, it assumes a runaway loop and mutes the port to protect your system. In this setup, it is almost always a false alarm triggered by Pro Tools sending an initial burst of data during the HUI handshake.

To unmute immediately

35. Open the loopMIDI window.
36. Click the port shown in red or orange to select it.
37. Click the Reset button at the bottom of the window.

To prevent it happening again

- In loopMIDI's Options menu, look for a feedback detection threshold setting or a "Do not auto-mute" option and disable it.

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- If your version of loopMIDI does not have this option, the muting will typically only occur once during the initial handshake. After resetting, it usually does not repeat during normal operation.
- Make sure Pro Tools does not have MIDI Thru or any echo enabled on the HUI controller ports.

Port numbers have changed

On Windows, the number appended to each MIDI port name can change if you plug or unplug USB devices, restart loopMIDI, or change the order in which devices are powered on. If a program stops connecting after a hardware change, run list_midi_ports.py to find the current numbers and update the configuration via Menu → Configure.

The HUI goes offline after a few seconds

The HUI expects a ping message at least once every two seconds to stay online. If HUI-Test is connected but the faders stop responding, check that the Auto-ping checkbox in the Connection tab is enabled. In HUI-Display, pings from Pro Tools are forwarded automatically as part of normal traffic and should not require intervention.

The program crashes on startup

If a program crashes before the window appears, the error details are written to a log file in the same folder as the program:

- `hui_display.py` → look for `hui_display_error.txt`
- `hui_test.py` → look for `hui_test_error.txt`

Alternatively, run the program from Command Prompt so the error stays visible on screen:

```
python hui_display.py
python hui_test.py
```

python-rtmidi installation fails (compiler error)

See Section 2.3, "If the installation fails". Installing the Microsoft C++ Build Tools and restarting resolves this in all known cases.

Chapter 7

Limitations

7.1 MIDI-Only Interface

HUI Tools communicates with the HUI hardware exclusively over MIDI. This is the same channel that Pro Tools uses, and it covers all of the normal operational functions of the unit — fader positions, button presses, LED states, VU meter levels, V-pot ring display, timecode display, and the main VFD. However, MIDI is a communications protocol, not a hardware diagnostics bus. There are several categories of HUI function that exist entirely within the unit's own electronics and firmware and cannot be observed or controlled over MIDI.

7.2 VFD Display Limitations

The main 2×40 vacuum-fluorescent display can only show content that has been sent to it over the MIDI connection as SysEx commands. HUI-Display intercepts and re-displays exactly what Pro Tools sends. However, there are two important cases where the VFD shows content that does not originate from the MIDI connection:

Internal ROM messages

Some text displayed on the VFD is generated by the HUI's own internal firmware, stored in ROM, without any corresponding MIDI command. This includes the startup splash screen shown when the unit powers on, and any status or diagnostic messages the firmware produces internally. These messages are not transmitted over MIDI and therefore cannot be captured or shown by HUI-Display.

Adjustment dialogs

Certain configuration dialogs on the HUI are handled entirely by the unit's own firmware. These dialogs appear on the VFD in response to hardware button combinations, but the text they display is generated locally in ROM and never sent over MIDI. As a result, HUI-Display cannot show them. Known examples include:

- **Talkback routing options** — Accessed from the Talkback button, this dialog allows routing of the talkback microphone signal. The options displayed are produced by the HUI's internal firmware.
- **Muting options** — The HUI's internal muting configuration dialog, accessed via specific button combinations, is similarly generated internally.
- **Dim attenuation** — The attenuation level for the monitoring DIM function is set through a firmware dialog that writes to the VFD without a MIDI command.

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> **NOTE:** If you need to access these dialogs, they will function normally on the physical VFD if it is working. HUI-Display does not interfere with these operations — it simply cannot see or reproduce them on the emulated display.

7.3 Hardware-Level Diagnostics

HUI-Test can only test what is reachable over MIDI. If a hardware fault exists below the MIDI layer — for example, a failed LED driver IC that prevents a whole zone of LEDs from lighting even though the MIDI command was received correctly, a VFD filament failure, or a power supply issue affecting only certain circuits — HUI-Test will not be able to detect it. In such cases:

- A command sent by HUI-Test may appear to succeed (no error is returned over MIDI) while the physical hardware produces no visible result.
- The Live View may show a button as transmitting correctly while the LED for that button fails to illuminate, because the LED and button circuits are independent.
- VU meter sliders set in HUI-Test confirm that the MIDI command was sent, but cannot confirm whether the meter LED segments are physically lit.

When using HUI-Test to diagnose hardware, always observe the physical unit directly alongside the software display. A discrepancy between what the software sent and what the hardware shows points to a hardware fault below the MIDI level.

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Chapter 8

Licensing

HUI Tools Software Licence

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Third-Party Acknowledgements

This software uses the following third-party libraries:

Library	Details

mido	MIDI Objects for Python. MIT Licence. https://mido.readthedocs.io/
python-rtmidi	Python binding for RtMidi. MIT Licence. https://github.com/SpotlightKid/python-rtmidi
tkinter	Standard Python GUI library. Part of the Python standard library (PSF Licence).

HUI Protocol

The HUI MIDI protocol implemented in this software is based on the reverse-engineering document "MACKIE HUI MIDI protocol" by theageman (2010), published by SSEI (www.ssei-online.de). That document is provided for informational purposes only, without warranty.

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